

Draft - expert review pending

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Topic 4.2 - OSH management systems

A real classroom situation

After the 2012 fire, a Karachi factory owner told a reporter, "We had safety measures. We had a fire extinguisher." He was not lying, and that is the point of this topic. An extinguisher on a wall is not a system. A system is what decides, before any fire, who checks the extinguisher, whether the alarm works, whether the doors are unlocked during shifts, and what happens to a manager who lets any of those slide.

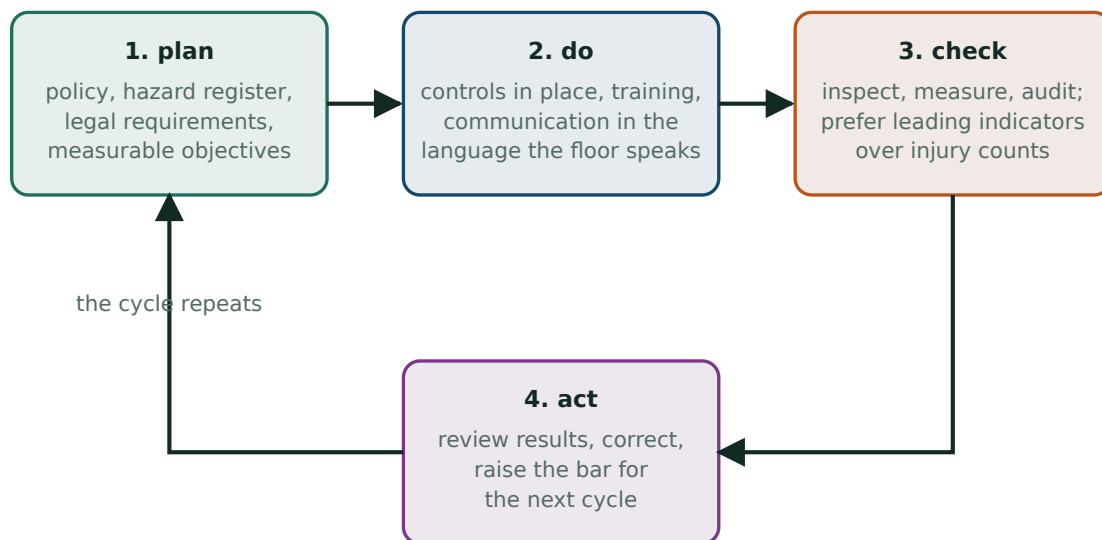
Explanation

Occupational safety and health management systems (international and national)

An OSH management system is the organised set of elements, policy, organisation, planning, procedures, checking and review, through which a workplace manages its hazards continuously rather than after each disaster. The international reference points are the ILO's Convention No. 155 on occupational safety and health (1981), which obliges member states to build national policy and enforcement, and ISO 45001:2018, the management-system standard workplaces can be certified against. Pakistan's national framework rests on its factory and labour legislation and its provincial enforcement, strengthened in recent decades by the ILO's post-fire programmes.

The engine of any of these systems is the same loop, borrowed from quality management: plan, do, check, act, PDCA.

The PDCA engine of every OSH system



Where the loop runs, the extinguisher on the wall is backed by a system; where it does not, it is paint.

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- **Plan:** write the policy, identify the hazards (Unit 3's walk and register), list the legal requirements, and set measurable objectives, for example, "all exits unlocked during every shift, checked hourly".
- **Do:** put the controls in place with the hierarchy from Unit 3, train the workers, and communicate the rules in a language the floor actually speaks.
- **Check:** inspect, measure, audit. Leading indicators, inspections completed, near-misses reported, count what the system is doing; lagging indicators, injuries and fatalities, count what already went wrong. A system steered only by injury counts is steered by its failures.
- **Act:** management reviews the results, corrects what the checking found, and raises the bar for the next cycle.

The two international reference points

Reference point	What it is	Who it binds	One contribution
ILO Convention	No. 155 (1981)	member states: national policy and enforcement	the state duty to build the enforcement system
ISO 45001:2018	the management-system standard	workplaces seeking certification	the PDCA system any workplace can run

Pakistan's national framework rests on factory and labour legislation with provincial enforcement.

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A worked example. Run PDCA on the burned factory, backwards, to see what was missing. Plan: no policy, no objectives, no legal check, so no unlocked-doors rule. Do: an extinguisher, but no training and no alarm drill. Check: no inspection had walked the escape routes for years. Act: no review had ever asked why previous small fires were extinguished and forgotten. Each element's absence is ordinary; the fire was their sum.

Common misconception. "A management system is paperwork for auditors." The paperwork is the residue, not the system. The system is the unlocked door at 3 p.m., the near-miss report that reaches the manager the same day, and the review that changes a rule because the numbers moved. Where those things happen, the paperwork describes them; where they do not, no binder of documents is a system.

Activity: PDCA on your school

Work in pairs. Take one hazard from your Unit 3 school walk, the loose wire or the open tank, and run the full PDCA cycle on it: write the objective, name the control and its hierarchy level, define the check with a leading indicator, and

state what the end-of-term review would read. Then swap with another pair and mark one gap in theirs. Allow 20 minutes.

Check your understanding

1. Name the four PDCA stages and one element of each. *(Remember)*
2. Distinguish leading from lagging indicators, with one example of each. *(Understand)*
3. Why is a system steered only by injury counts "steered by its failures"? *(Analyze)*
4. Your school has a fire drill but no alarm test schedule. Which PDCA stage is weakest, and what would you add? *(Apply)*

Summary

An OSH management system manages hazards continuously through the PDCA loop: plan the policy, hazards and objectives; do the controls, training and communication; check with inspection, measurement and audit, preferring leading indicators; act on review and correction. The ILO's Convention 155 and ISO 45001:2018 are the international reference points, with national frameworks resting on factory and labour law. The extinguisher on the wall without the loop around it is what the Karachi fire already taught.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the four PDCA stages and their elements.
- ☐ I can give one leading and one lagging indicator for the same hazard.
- ☐ I can run PDCA on a hazard from my own school walk.
- ☐ I can explain why documents alone are not a system.

Try this at your practicum school

Ask whether the school has a fire drill, and then ask the caretaker when the alarm was last tested and by whom. Write three sentences: what exists, what is missing, and which PDCA stage the missing piece belongs to. Do not name the school.

Summative task

Write 150 to 200 words on the following.

A school owner says, "We have fire extinguishers, so we are safe." Write the reply this topic supports: what a system adds to equipment, using the PDCA stages, and the two cheapest elements the school could add this month.

Mini-rubric

Criterion	What a strong response does
System explained (Understand)	Contrasts equipment with the loop, not just listing stages
Stages used (Apply)	Names what each stage would add for a school fire risk
Cheapest-first (Evaluate)	Two realistic, near-zero-cost elements
Clarity and register	Plain sentences for an owner; within the word count

Further reading

- World Health Organization. *Occupational health* (health topic): <https://www.who.int/health-topics/occupational-health>
- Harrington, J. M., & Gill, F. S. (Eds.). *Occupational Health*. Blackwell. (guide-recommended book, cited by reference only)
- Shilling, R. S. F. *Occupational Health Practice*. (guide's "Environmental Health Practice" closest real work, cited by reference only; flagged per D-

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